

Who Defends Whom?

From Argument Acceptability to Social Conflict

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What Questions Does This Talk Address?

Social network research often focuses on positive relations, such as friendship, kinship, cooperation, and communication. Having discussed these relations, we now turn to negative relations: how do conflict, hostility, and exclusion shape actors and their groups?

Since networks of negative social relations and argumentation frameworks both have graph structures, we use extensions to select and interpret actor groups.

The First Obstacle: Undirected Conflict and Trivial Self Defence

How can we rule out trivial self defence and identify effective defence in undirected conflict networks?

Building on this question, we consider

1. how external defence can explain groups in negative social networks;
2. which actors or groups can serve as conflict brokers when positive and negative relations coexist;
3. how actor-level social relations affect argument-level defence and acceptability.

Outline

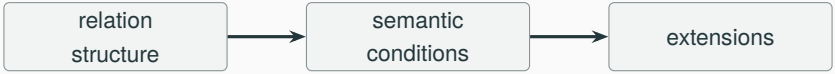
Part I	External Defence in Argumentation Frameworks
Part II	From Argument Extensions to Actor Groups
Part III	Bipolar Social Networks and Conflict Brokerage Semantics
Part IV	Open Questions on Actor–Argument Interaction

Main Idea

Can we bring extension-based analysis from abstract argumentation to social networks and develop semantics suited to their own problems? When arguments are put forward by actors in a social network, how does interaction between the two levels affect conflict, defence, and acceptability?

Why Draw on Argumentation Semantics?

Argumentation semantics selects extensions through conditions on graph structure.



For Social Networks

We use extensions to identify actor groups with specific social roles and develop semantics tailored to social network problems.

An Argumentation Perspective on Social Conflict

Negative social relation

$$aR^{-}b$$

Conflict relations in social networks are usually undirected.



Symmetric attacks

$$a \rightharpoonup b \quad \text{iff} \quad b \rightharpoonup a$$

In an abstract argumentation framework, this becomes symmetric attack.

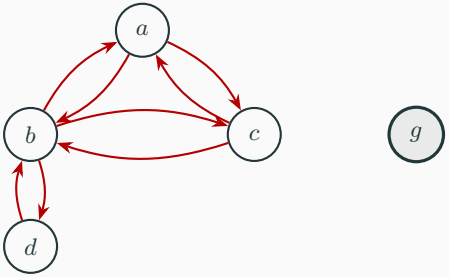


Note

In a symmetric framework, every argument can defend itself against all its attackers, so every conflict-free set is admissible. What should count as effective defence in symmetric conflict?

External Defence in Argumentation Frameworks

A Dilemma in Symmetric Argumentation Frameworks



Dung admissibility

Every attack allows the attacked argument to counterattack. Thus, in symmetric frameworks, conflict-free sets and admissible sets coincide:

$$ADF(SAF) = CF(SAF)$$

Strong admissibility

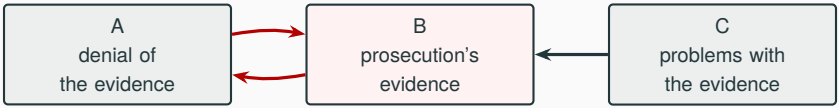
Defence chains must start from unattacked arguments. In symmetric frameworks, only isolated arguments can belong to strongly admissible sets.

Dung admissibility is too permissive; strong admissibility is too restrictive.

Motivation in General Frameworks: Courtroom Defence

External defence also applies beyond symmetric frameworks

- In the O.J. Simpson trial example, A represents a denial of the prosecution's evidence.
- B is the prosecution's evidence and directly attacks A.
- C identifies problems with that evidence and provides third-party defence for A.



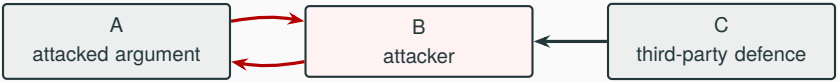
Formal reading

$$B \rightharpoonup A, \qquad A \rightharpoonup B, \qquad C \rightharpoonup B, \qquad C \neq A.$$

A's response to B is still trivial self defence; external backing comes from C, a distinct argument.

The Basic Idea of External Defence

- A is attacked by B.
- A's counterattack against B is still only trivial self defence.
- A distinct argument C attacking B provides third-party defence.



$$B \rightharpoonup A, \qquad A \rightharpoonup B, \qquad C \rightharpoonup B, \qquad C \neq A$$

Defining External Defence

For an argumentation framework $AF = \langle Arg, \rightharpoonup \rangle$ and a set $\Delta \subseteq Arg$:

External defence

$$\Delta \text{ } \epsilon\text{-defends } C \quad \text{iff} \quad \forall B,\ B \rightharpoonup C \Rightarrow \Delta \setminus \{C\} \rightharpoonup B.$$

External defence function

$$d_\epsilon(\Delta) = \{C \in Arg \mid \Delta \text{ } \epsilon\text{-defends } C\}.$$

External admissibility

Δ is ϵ-admissible iff it is conflict-free and

$$\Delta \subseteq d_\epsilon(\Delta).$$

Three Notions of Admissibility

Admissibility	Intuition	Requirement
Dung admissibility	Allows arguments to defend themselves.	The set is conflict-free, and each member is defended by Δ.
External admissibility	Requires third-party defence from another argument in the admissible set.	The set is conflict-free, and each member A is defended by $\Delta \setminus \{A\}$.
Strong admissibility	Requires defence chains to be traced back to unattacked arguments.	Each argument in the admissible set has a finite defence chain rooted in unattacked arguments.

External admissibility rules out trivial self defence without requiring an unattacked starting point.

Where External Admissibility Fits

A spectrum of admissibility

$$sADF(AF) \subsetneq \epsilon\text{-}ADF(AF) \subsetneq ADF(AF)$$

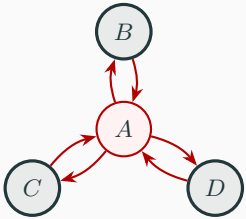
- Dung admissibility allows arguments to defend themselves.
- External admissibility requires third-party defence.
- Strong admissibility requires defence chains starting from unattacked arguments.

Interpretation

External admissibility lies between Dung admissibility and strong admissibility: it is more restrictive than the former and less restrictive than the latter.

Two-Node Mutual Attack, Four-Cycles, and Stars

A star



Any two leaves can defend each other against attacks from the centre A.

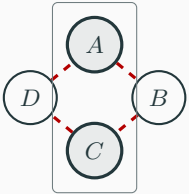
Two comparisons

Mutual attack



Neither $\{A\}$ nor $\{B\}$ is accepted.

A four-cycle



$\{A, C\}$ and $\{B, D\}$ can be accepted.

What matters is whether the counterattack comes from a member other than the attacked argument.

Why

Two-node mutual attack offers only trivial self defence. Stars and four-cycles show how group members can provide third-party defence for one another.

External Semantics

Extensions constructed from d_ϵ

- Δ is **ϵ-complete** iff Δ is conflict-free and $\Delta = d_\epsilon(\Delta)$.
- Δ is **ϵ-preferred** iff Δ is an inclusion-maximal ϵ-admissible set.
- Δ is **ϵ-grounded** iff Δ is the least fixed point of d_ϵ.

Basic results

- Every finite argumentation framework has an ϵ-preferred extension.
- The ϵ-grounded extension coincides with Dung's grounded extension.
- Stable extensions remain unchanged, but are no longer guaranteed to be ϵ-preferred.

Part I: Summary

- Negative social relations naturally induce symmetric argumentation frameworks.
- In these structures, Dung admissibility collapses to conflict-freeness.
- Strong admissibility excludes most nodes involved in conflict.
- External defence rules out trivial self defence while preserving third-party and collective defence.

Towards Part II

How should these externally admissible sets be interpreted in social networks?

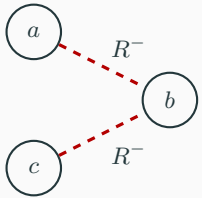
From Argument Extensions to Actor Groups

The Framework Induced by Negative Social Relations

Negative social network

$$\mathcal{N}^- = \langle A, R^- \rangle$$

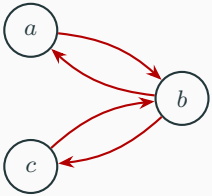
R^- represents conflict, hostility, exclusion, or opposition between actors.



Induced symmetric AF

$$AF^-(\mathcal{N}) = \langle A, \rightharpoonup \rangle, \qquad a \rightharpoonup b \quad \text{iff} \quad aR^-b.$$

Since R^- is undirected, each negative edge induces a pair of mutual attacks.



The graph is unchanged; what we interpret has changed.

When nodes represent actors rather than arguments, how should we interpret an extension?

A Social Reading of Extensions

	Abstract argumentation	Social network reading
Nodes	arguments	actors
Relations	attacks	negative social relations
Extensions	acceptable arguments	actor groups with a structural role
Focus	Which arguments can be accepted together?	Which actor groups can jointly face external conflict?

A Shift in Research Focus

We draw on the method of semantic extensions to explain the roles of groups in social networks.

Alliance Group

For $S \subseteq A$, regard S as a candidate group of actors.

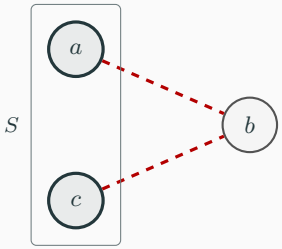
Conflict-free

$$\neg\exists a, c \in S : \ aR^{-}c.$$

External defence

$$\forall a \in S, \ \forall b \notin S, \ bR^{-}a \Rightarrow \exists c \in S \setminus \{a\} : cR^{-}b.$$

Alliance groups and externally admissible sets share the same topological requirements; their intuitions come from social networks and abstract argumentation frameworks, respectively.

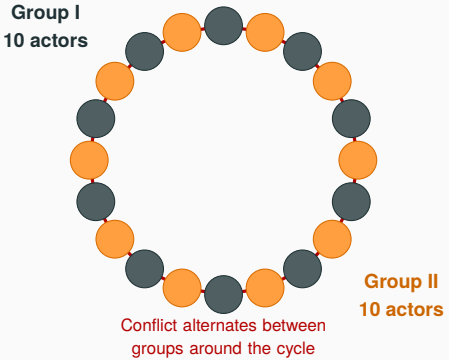


When a faces an external opponent b, c can act as a third-party defender within the group.

Intuition in Social Networks

An alliance group has the structure of an externally admissible set: whenever a member faces external conflict, at least one ally stands alongside them.

From Even Cycles to Group Opposition



Scaling up the four-cycle

In a larger even cycle, actors from two sides alternate. Each side has no internal negative conflict, yet repeatedly conflicts with the other side. Each side may then receive external defence.

This suggests a natural interpretation:

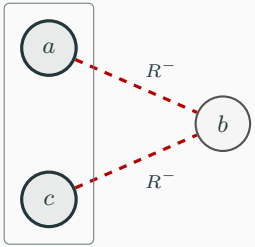
$$\text{within-group backing} \; + \; \text{between-group conflict}.$$

Note

This does not define polarization semantics; it suggests that external semantics can capture a structure resembling polarization.

Alliance and R^+

Alliance uses negative relations alone



Clarifying the interpretation

Here, saying that c is allied with a means:

$$\text{when } aR^-b, \text{ there is } c \text{ such that } cR^-b$$

That is, c can also respond to a's external opponent b.

This alliance-like relation does not assert that

$$aR^+c \quad \text{or} \quad cR^+a.$$

Why Move to the Next Part?

Positive social relations must be introduced separately as R^+. In a bipolar social network, this raises a new semantic question: which actors can access both sides of a conflict?

Part II: Summary

- Negative social relations can induce symmetric argumentation frameworks.
- The interpretation of extensions shifts from acceptable arguments to actor groups with social roles.
- Externally admissible groups can be understood as alliance-like groups.
- Even cycles and larger alternating conflict structures can exhibit within-group backing and between-group conflict.

Towards Part III

So far we have used negative relations alone. We now add R^+ to study a role specific to social networks: conflict brokerage.

Bipolar Social Networks and Conflict Brokerage Semantics

Bipolar Social Networks

Bipolar social network

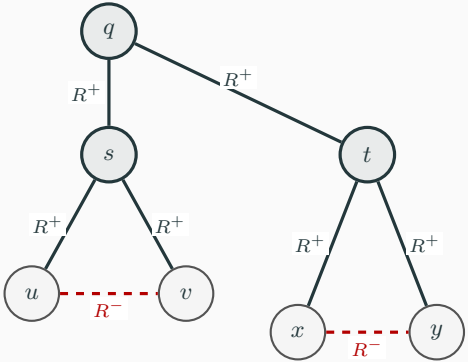
$$\mathcal{N} = \langle A, R^+, R^- \rangle$$

R^- **negative social relation**
Conflict, hostility, opposition, and sanctions.

R^+ **positive social relation**
Friendship, group membership, social interaction, and cooperation.

The Question for Part III

When a social network contains both R^- and R^+, can the extension-based approach of abstract argumentation guide the development of semantics specifically for social networks?



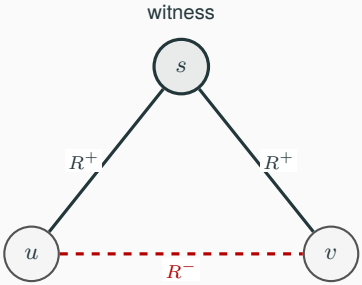
Brokerage witness

The basic structure

$$uR^{-}v, \qquad sR^{+}u, \qquad sR^{+}v$$

When u and v have a negative relation and s has positive access to both:

s witnesses the conflict $\{u, v\}$.

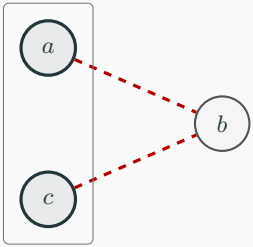


Note

"Conflict brokerage" means having access to both parties to a conflict; it does not presuppose the ability to resolve it.

Two Different Questions

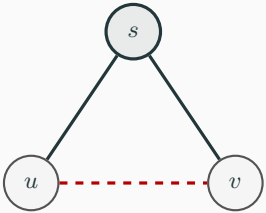
External Semantics



Who stands alongside group members when they face external conflict?

Here, alliance still arises through R^-: another member responds to the same external opponent.

Conflict Brokerage Semantics



Who can access both sides of an external conflict?

Here, R^+ is central: an actor has positive access to both endpoints of the conflict.

A Shift in Research Focus

Conflict brokerage applies extension-based methods to roles specific to social networks; it is more than adding positive edges to external semantics.

Brokerage range

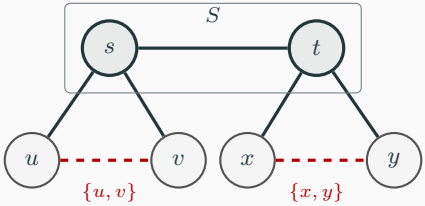
For a set of actors $S \subseteq A$ connected by R^+, consider its brokerage range $Br_w(S)$:

$$Br_w(S) = \big\{\{u, v\} \mid u, v \in A \setminus S,\ uR^-v,\ \exists s \in S :\ sR^+u \wedge sR^+v\big\}.$$

Interpretation

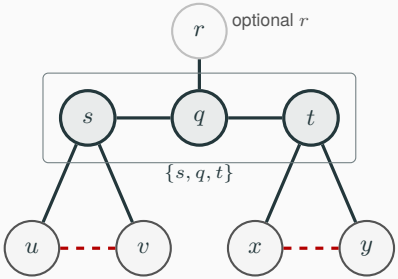
$Br_w(S)$ records which external conflicts S covers, rather than which actors belong to S.

This lets us compare the conflict brokerage roles of different groups S.



Candidate group and brokerage range

A candidate group



$\{s, q, t\}$ is internally connected by positive relations and has access to two external conflicts.

Comparing conflict brokerage ranges

Candidate S	$Br_w(S)$	Social reading
$\{s\}$	$\{\{u, v\}\}$	s witnesses the conflict between u, v
$\{t\}$	$\{\{x, y\}\}$	t witnesses the conflict between x, y
$\{s, q, t\}$	$\{\{u, v\}, \{x, y\}\}$	q connects the two witnesses into a group
$\{s, q, t, r\}$	$\{\{u, v\}, \{x, y\}\}$	The added candidate actor r witnesses no new conflict

Here we compare ranges $Br_w(S)$ only; next we ask how to select S from candidate groups based on $Br_w(S)$.

Basic and free-rider refinement

Basic level

At this stage, S is maximal: adding another candidate reveals a previously unwitnessed conflict.

The basic level focuses on coverage and may retain redundant members.

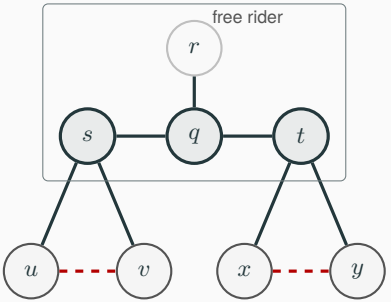
Then, exclude free riders

$$Br_w(\{s, q, t, r\}) = Br_w(\{s, q, t\})$$

A member is unnecessary for brokerage if its removal preserves feasibility and leaves Br_w unchanged.

Further level

No proper subset preserves the same Br_w (excluding all free riders).



r is connected to the group but witnesses no additional external conflict.

Part III: Summary

- Bipolar social networks contain both R^- and R^+.
- R^+ represents social access, not argument-level support.
- Conflict brokerage concerns who can access both endpoints of an external conflict.
- $Br_w(S)$ captures the external conflicts covered by a group.
- The further refinement excludes free riders whose removal leaves the brokerage range unchanged.

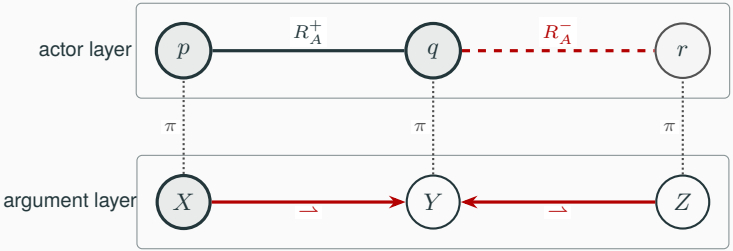
Towards Part IV

We now bring the two levels together: when arguments are put forward by particular actors, how do actor-level social relations affect argument-level acceptability?

Open Questions on Actor–Argument Interaction

Connecting the Two Levels

External defence is defined at the argument level; conflict brokerage is defined at the actor level. In practice, however, arguments are usually put forward by particular actors, who in turn have social relations with one another.



The Question for Part IV

When we associate arguments with their proponents, how do social relations affect defence, conflict, and acceptability?

A Two-Level Representation for Discussion

$$M = \langle A, Arg, \pi, R_A^+, R_A^-, \rightharpoonup \rangle$$

A	actors
Arg	arguments
$\pi : Arg \to A$	maps each argument to its proponent or author
R_A^+, R_A^-	actor-level positive and negative social relations
$\rightharpoonup$	argument-level attack relation

Note

This representation is intended to frame an open discussion. It is not yet a completed formal model with established properties.

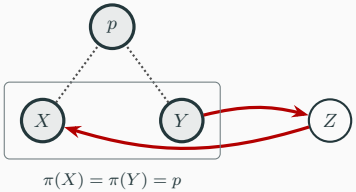
Open Question 1: Levels of Third-Party Defence

What external defence requires

At the argument level, the defender must be a distinct argument:

$$X \neq Y.$$

Yet a distinct argument need not correspond to a distinct third party.

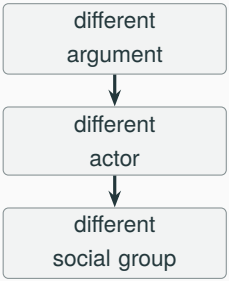


Question

If two arguments come from the same actor, is argument-level externality sufficient?

Open Question 1: A Third Party in the Social Sense

A conceptual progression



This is a conceptual progression to be clarified, rather than a theorem.

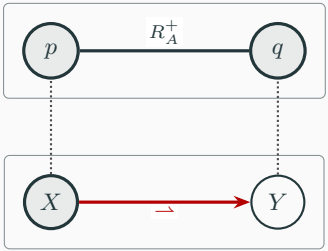
The connection to polarization

Even if $\pi(X) \neq \pi(Y)$, the two proponents may belong to the same closely aligned camp.

At the argument level, each side may receive ample external defence. At the social level, all corroboration may still come from within the same camp.

External semantics is not itself a semantics of polarization, but it permits an interpretation in terms of within-group backing and between-group conflict.

Open Question 2: Attacks within Positive Relations



The same attack in a different social context

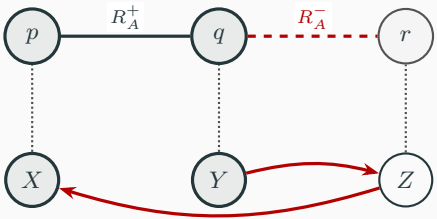
Suppose pR_A^+q, but X, put forward by p, attacks Y, put forward by q. Could we treat this as an internal conflict and thereby weaken the force of $X \rightharpoonup Y$? Should $X \rightharpoonup Y$ receive exactly the same interpretation as an attack between arguments whose proponents satisfy pR^-q?

A Cautious Formulation

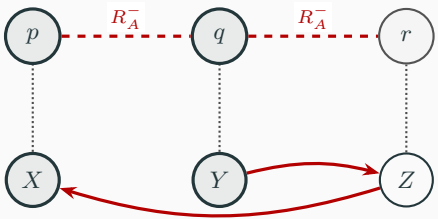
The attack may retain the same logical structure while occurring in a different social context.

Open Question 2: Social Relations and Interpreting Defence

The same argument pattern can be embedded in different social relations



Y defends X, and p and q have a positive relation.



The defence pattern is the same, but the proponents have a different social relation.

Question

Should social relations affect the independence, explanatory force, or social credibility of defence?

Open Question 3: Brokerage of Argumentative Conflict

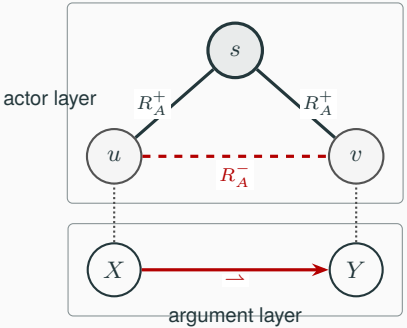
Starting from actor-level brokerage

In Part III, s has access to both u and v, with uR_A^-v.

Now consider conflict at the argument level:

$$X \rightharpoonup Y, \qquad \pi(X) = u, \quad \pi(Y) = v.$$

If sR_A^+u and sR_A^+v, might s have an argumentative brokerage role?



Open Question 3: From Br_w to Argumentative Conflict

Brokerage of argument-level conflict

Suppose an actor s has access to two proponents:

$$sR_A^+\pi(X), \qquad sR_A^+\pi(Y),$$

and their arguments are in conflict:

$$X \rightharpoonup Y,$$

Then s may also have an **argumentative brokerage role**.

One could tentatively define:

$$Br_{w,Arg}(s) = \{(X, Y) \mid X \rightharpoonup Y, \\ sR_A^+\pi(X),\ sR_A^+\pi(Y)\}.$$

Through s's social access, argument-level conflict falls within its brokerage scope.

Is this a form of "internal conflict"?

It can be viewed as **argumentative conflict within the brokerage scope**: the conflict occurs at the argument level, while both proponents are within s's social reach.

Comparison with Actor-Level Admissibility

Actor-level admissibility asks whether a group is conflict-free and can defend its members against external attacks.

Who Defends Whom

The central question of this research programme



The Question That Remains

Who defends whom depends on the argument graph, on who puts forward the arguments, on their social relations, and on who has access to whom.

- What counts as a third party?
- How do social relations affect argument-level conflict and defence?
- How does actor-level brokerage connect to argument-level acceptability?